

OS & SP

Name : M Ramya Sree

Roll No : 2520030094

Practical

1.Virtual Address

```
GNU nano 8.7.1
#include <stdio.h>
int main(){
int x = 10;
printf("Value of x = %d\n", x);
printf("Virtual address of x = %p\n", (void *)&x);
return 0;
}
```

```
ramyasree@DESKTOP-DP209U6:~$ nano virtualaddress.c
ramyasree@DESKTOP-DP209U6:~$ gcc virtualaddress.c -o virtualaddress
ramyasree@DESKTOP-DP209U6:~$ ./virtualaddress
Value of x = 10
Virtual address of x = 0x7ffd068f7d04
ramyasree@DESKTOP-DP209U6:~$ |
```

2. Address Space

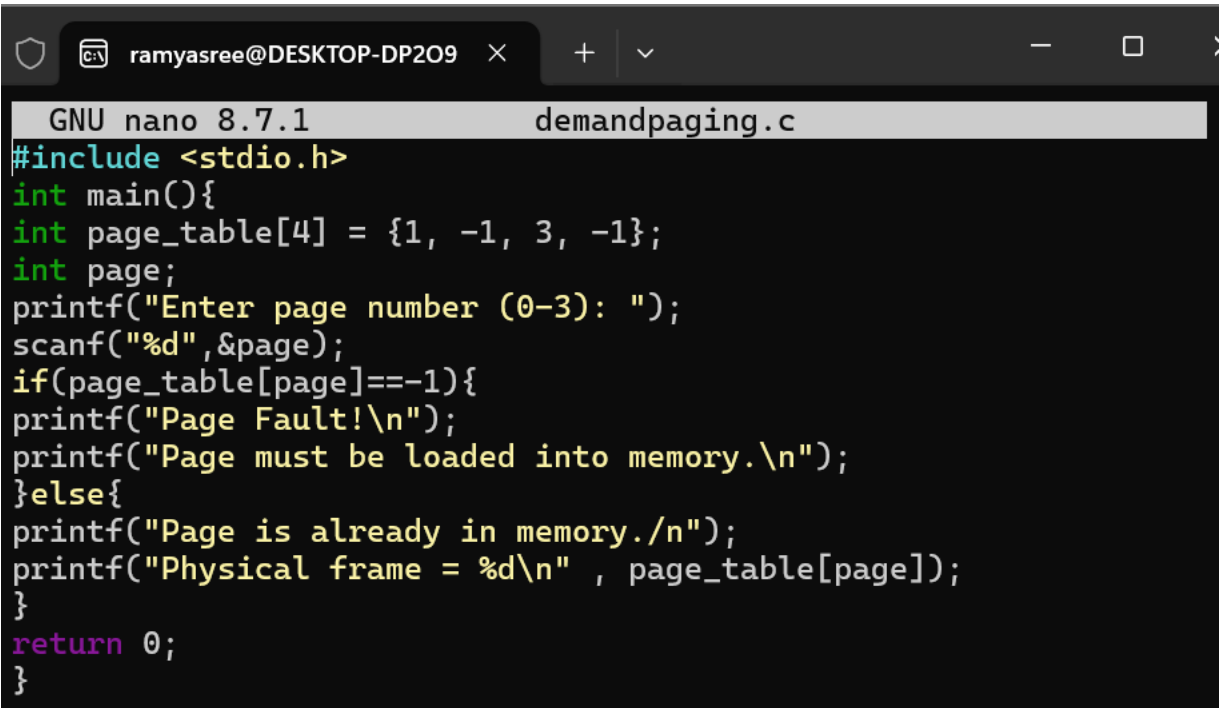
```
GNU nano 8.7.1
#include<stdio.h>
#include<stdlib.h>
int global = 10;
int bss_variable;
int main(){
int local = 20;
int*heap = malloc(sizeof(int));
*heap = 30;
printf("Code address : %p\n", (void *) main);
printf("Data address : %p\n", (void *) &global);
printf("BSS address : %p\n", (void *)&bss_variable);
printf("Heap address : %p\n", (void *)heap);
printf("Stack address : %p\n", (void *)&local);
free (heap);
return 0;
}
```

```

ramyasree@DESKTOP-DP209U6:~$ nano addressspace.c
ramyasree@DESKTOP-DP209U6:~$ gcc addressspace.c -o addressspace
ramyasree@DESKTOP-DP209U6:~$ ./addressspace
Code address : 0x5e5aab68b1a9
Data address : 0x5e5aab68e010
BSS address : 0x5e5aab68e018
Heap address : 0x5e5acbb90010
Stack address : 0x7fff7b862fbc
ramyasree@DESKTOP-DP209U6:~$ |

```

3. Demand Paging



The screenshot shows a web browser window with a single tab titled 'ramyasree@DESKTOP-DP209'. The browser's address bar shows 'GNU nano 8.7.1' and the file 'demandpaging.c'. The code in the editor is as follows:

```

#include <stdio.h>
int main(){
int page_table[4] = {1, -1, 3, -1};
int page;
printf("Enter page number (0-3): ");
scanf("%d",&page);
if(page_table[page]==-1){
printf("Page Fault!\n");
printf("Page must be loaded into memory.\n");
}else{
printf("Page is already in memory.\n");
printf("Physical frame = %d\n" , page_table[page]);
}
return 0;
}

```

```

ramyasree@DESKTOP-DP209U6:~$ nano demandpaging.c
ramyasree@DESKTOP-DP209U6:~$ gcc demandpaging.c -o demandpaging
ramyasree@DESKTOP-DP209U6:~$ ./demandpaging
Enter page number (0-3): 2
Page is already in memory.\nPhysical frame = 3
ramyasree@DESKTOP-DP209U6:~$ |

```

4. Virtual Physical Address

```
GNU nano 8.7.1 virtualphysicaladdress.c
#include <stdio.h>
int main(){
int page_table[] = {3, 5, 1, 7};
int page_size = 100;
int virtual_address = 250;
int page = virtual_address / page_size;
int offset = virtual_address % page_size;
int frame = page_table[page];
int physical_address = frame * page_size + offset;
printf("Virtual Address = %d\n", virtual_address);
printf("Page Number = %d\n", page);
printf("Offset = %d\n", offset);
printf("Frame Number = %d\n", frame);
printf("Physical Address = %d\n", physical_address);
return 0;
}
```

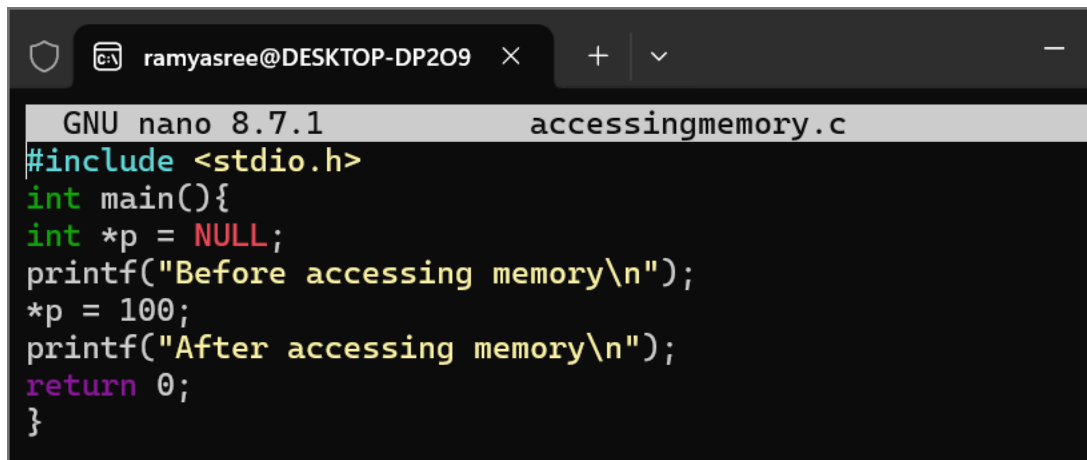
```
ramyasree@DESKTOP-DP209U6:~$ nano virtualphysicaladdress.c
ramyasree@DESKTOP-DP209U6:~$ gcc virtualphysicaladdress.c -o virtualphysicaladdress
ramyasree@DESKTOP-DP209U6:~$ ./virtualphysicaladdress
Virtual Address = 250
Page Number = 2
Offset = 50
Frame Number = 1
Physical Address = 150
ramyasree@DESKTOP-DP209U6:~$
```

5. Memory Allocation

```
GNU nano 8.7.1 memoryallocation.c
#include <stdio.h>
#include <stdlib.h>
int main() {
int *p;
p = malloc(5*sizeof(int));
if(p==NULL) {
printf("Memory allocation failed\n");
return 1;
}
for(int i = 0; i<5; i++){
p[i]=(i+1)*10;
}
for(int i=0; i<5; i++) {
printf("%d", p[i]);
}
printf("/n");
free(p);
return 0;
}
```

```
ramyasree@DESKTOP-DP209U6:~$ nano memoryallocation.c
ramyasree@DESKTOP-DP209U6:~$ gcc memoryallocation.c -o memoryall
ocation
ramyasree@DESKTOP-DP209U6:~$ ./memoryallocation
1020304050/nramyasree@DESKTOP-DP209U6:~$ |
```

6. Accessing Memory



```
GNU nano 8.7.1 accessingmemory.c
#include <stdio.h>
int main(){
int *p = NULL;
printf("Before accessing memory\n");
*p = 100;
printf("After accessing memory\n");
return 0;
}
```

```
ramyasree@DESKTOP-DP209U6:~$ nano accessingmemory.c
ramyasree@DESKTOP-DP209U6:~$ gcc accessingmemory.c -o accessingm
emory
ramyasree@DESKTOP-DP209U6:~$ ./accessingmemory
Before accessing memory
Segmentation fault ./accessingmemory
ramyasree@DESKTOP-DP209U6:~$ |
```